



ADER Methods for Hyperbolic Equations with a Time-Reconstruction Solver for the Generalized Riemann Problem: the Scalar Case

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Abstract

The ADER approach to solve hyperbolic equations to very high order of accuracy has seen explosive developments in the last few years, including both methodological aspects as well as very ambitious applications. In spite of methodological progress, the issues of efficiency and ease of implementation of the solution of the associated generalized Riemann problem (GRP) remain the centre of attention in the ADER approach. In the original formulation of ADER schemes, the proposed solution procedure for the GRP was based on (i) Taylor series expansion of the solution in time right at the element interface, (ii) subsequent application of the Cauchy–Kowalewskaya procedure to convert time derivatives to functionals of space derivatives, and (iii) solution of classical Riemann problems for high-order spatial derivatives to complete the Taylor series expansion. For realistic problems the Cauchy–Kowalewskaya procedure requires the use of symbolic manipulators and being rather cumbersome its replacement or simplification is highly desirable. In this paper we propose a new class of solvers for the GRP that avoid the Cauchy–Kowalewskaya procedure and result in simpler ADER schemes. This is achieved by exploiting the history of the numerical solution that makes it possible to devise a time-reconstruction procedure at the element interface. Still relying on a time Taylor series expansion of the solution at the interface, the time derivatives are then easily calculated from the time-reconstruction polynomial. The resulting schemes are called ADER-TR. A thorough study of the linear stability properties of the linear version of the schemes is carried out using the von Neumann method, thus deducing linear stability regions. Also, via careful numerical experiments, we deduce stability regions for the corresponding non-linear schemes. Numerical examples using the present simplified schemes of fifth and seventh order of accuracy in space and time show that these compare favourably with conventional ADER methods. This paper is restricted to the one-dimensional scalar case with source term, but preliminary results for the one-dimensional Euler equations indicate that the time-reconstruction approach offers significant advantages not only in terms of ease of implementation but also in terms of efficiency for the high-order range schemes.

Keywords Hyperbolic equations · Finite volume · ADER methods · Generalized Riemann problem (GRP) · Time-reconstruction (TR)

1 Introduction

We are interested in solving numerically time-dependent partial differential equations of the form

$$\partial_t \mathbf{Q}(x, t) + \partial_x \mathbf{F}(\mathbf{Q}(x, t)) = \mathbf{S}(\mathbf{Q}(x, t)), \quad x \in \Omega, \quad t > 0 \quad (1)$$

with appropriate initial and boundary conditions. Here $\mathbf{Q}(x, t)$ is the vector of unknowns; $\mathbf{F}(\mathbf{Q}(x, t))$ and $\mathbf{S}(\mathbf{Q}(x, t))$ are, respectively, the vectors of fluxes in one space dimension and the vector of source terms, which are prescribed functions of the unknown. In this paper we focus on the scalar one-dimensional case with a source term

$$\partial_t q(x, t) + f(q(x, t)) = s(q(x, t)), \quad x \in \Omega, \quad t > 0. \quad (2)$$

Exact integration of (2) in the space–time control volume $[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \times [0, \Delta t]$ gives

$$\hat{q}_i^{n+1} = \hat{q}_i^n - \frac{\Delta t}{\Delta x} \left(\hat{f}_{i+\frac{1}{2}} - \hat{f}_{i-\frac{1}{2}} \right) + \Delta t \hat{s}_i, \quad (3)$$

where $x_{i+\frac{1}{2}}$ denotes the intercell boundary of the computational cell $[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$; Δt is the time step and

$$\begin{cases} \hat{q}_i^n = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} q(x, t^n) dx, \\ \hat{f}_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} f\left(q\left(x_{i+\frac{1}{2}}, t\right)\right) dt, \\ \hat{s}_i = \frac{1}{\Delta t \Delta x} \int_{t^n}^{t^{n+1}} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} s(q(x, t)) dx dt. \end{cases} \quad (4)$$

Formula (3), along with definitions (4), is exact and is the starting point for designing conservative, fully-discrete one-step finite-volume ADER schemes of the form

$$q_i^{n+1} = q_i^n - \frac{\Delta t}{\Delta x} \left(f_{i+\frac{1}{2}} - f_{i-\frac{1}{2}} \right) + \Delta t s_i, \quad (5)$$

as first proposed by Toro et al. [35] for linear problems in one and multiple space dimensions. Here $\{q_i^n\}$ denotes an integral cell average. The ADER schemes (5) result from re-interpreting (3, 4) in an approximate manner with numerical flux

$$f_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} f(q_{\text{LR}}(\tau)) d\tau \quad (6)$$

and numerical source

$$s_i = \frac{1}{\Delta t \Delta x} \int_{t^n}^{t^{n+1}} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} s(q_i(x, \tau)) dx d\tau \quad (7)$$

These approximations are constructed by finding an approximation $q_{LR}(\tau) \approx q(x_{i+\frac{1}{2}}, t)$ at the interface, as a function of time, and an approximation $q_i(x, \tau) \approx q(x, t)$ within the control volume, as a function of both space and time.

The ADER method has two main ingredients:

- i. high-order non-linear spatial reconstruction (once per time step and typically a WENO reconstruction [12, 18]) based on cell averages $\{q_i^n\}$ and
- ii. solution of the generalized Riemann problem, GRP, also called the high-order Riemann problem, that allows the approximate evaluation of integral (6) to the desired order of accuracy. We remark that the evaluation of the numerical source (7) is similar to that of the numerical flux (6).

Therefore, central to the ADER approach is the generalized Riemann problem, that is the special initial value problem

$$\begin{cases} \text{PDEs: } \partial_t q + \partial_x f(q) = s(q), & x \in (-\infty, \infty), \quad t > 0, \\ \text{ICs: } q(x, 0) = \begin{cases} q_L(x) & \text{if } x < 0, \\ q_R(x) & \text{if } x > 0. \end{cases} \end{cases} \quad (8)$$

Two features stand out in the generalized Riemann problem, as compared to the conventional piece-wise constant data Riemann problem, namely the initial conditions are piece-wise smooth (e.g., polynomial functions of any degree) and the source term is included. In what follows we denote by GRP_K the generalized Riemann problem with initial conditions that are smooth functions of x with K continuous non-trivial spatial derivatives.

For a description of the ADER approach see the original reference [35] and subsequent papers by Toro and Titarev [29, 37]. For useful elementary background see Chaps. 19 and 20 of the textbook [32]. For related works on theoretical aspects of the generalized Riemann problem see Glimm et al. [8], Tatsien and Wenci [27], LeFloch and Raviart [16] and LeFloch and Tatsien [17]. Regarding numerical schemes utilising aspects of the generalized Riemann problem, early works include that of Ben-Artzi and Falcovitz [2], who proposed a second-order Godunov-type method. See also Men'Shov [20] for a second-order method. Harten et al. [11] were credited with the first high-order Godunov method beyond second order of accuracy which employs Taylor series expansions within each element and use of the Cauchy–Kowalewskaya procedure. However, it is worth mentioning that in their original work the authors did not explicitly formulate their scheme in terms of the solution of the generalized Riemann problem at the element interface; this is, however, possible, as shown later by Castro and Toro [3], from which one may infer a solution method for the GRP_K , called by these authors as the HEOC generalized Riemann problem solver.

During the 1990s Toro and collaborators put forward the idea behind the ADER framework to construct high-order methods based on the solution of the generalized Riemann problem right at the interface, by extending the method of Ben-Artzi and Falcovitz [2] to any desired accuracy. These early ADER methods were first coherently communicated in [35] for linear problems in one and multiple space dimensions and implemented up to tenth order of accuracy in space and time. This was inspired by the LeFloch–Raviart Taylor series expansion and their GRP solver [16], and a simplification of the method of Ben-Artzi and Falcovitz [2] presented in Chap. 14 of [31]. The approximate semi-analytical solution is found right at the cell interface and the equations include source terms. The extension of the ADER framework

to non-linear systems [29] was possible after the GRP solution procedure proposed by Toro and Titarev [37]. Since then other methods of solution have been proposed. Examples include Castro and Toro [3], Dumbser et al. [7], Toro and Montecinos [36], Goetz and Iske [10] and Goetz and Dumbser [9]. Currently, the original method of Harten and collaborators [11] may be interpreted as an ADER scheme that extends the second-order MUSCL-Hancock method [41], with the HEOC solver for the GRP [3] or the solver of Dumbser et al. [7].

In this paper we propose a novel method for solving the generalized Riemann problem, retaining the LeFloch–Raviart Taylor series expansion [16]. The method is presented in the next section.

2 Time-Reconstruction Solvers for the GRP

In this section we present a new class of methods to compute the solution of the generalized Riemann problem at $x = 0$, the interface, as a function of time. The novel strategy proposed here differs from conventional solvers for the GRP_K (such as the Toro–Titarev [29, 33, 37], the HEOC method of Harten et al. [11] and the Castro–Toro [3] solver) in that it does not resort to the rather cumbersome Cauchy–Kowalewskaya procedure to express time derivatives as functions of space derivatives, which for realistic problems is only feasible by the use of symbolic manipulators. Simplifications are thus very attractive. We expect that the advantages afforded by this new approach will be significant for very complicated non-linear multidimensional problems. In this paper, however, we restrict our attention to the scalar case in one space dimension (2) to present the methodology.

Let us consider the GRP_K (8) at a given time level n . As usual, we seek an approximate solution $q_{\text{LR}}(\tau)$ of this problem at $x = 0$ as the time power series expansion

$$q_{\text{LR}}(\tau) = q(0, 0_+) + \sum_{k=1}^K \left[\partial_t^{(k)} q(0, 0_+) \right] \frac{\tau^k}{k!}, \quad (9)$$

where the leading term of the expansion is

$$q(0, 0_+) = \lim_{t \rightarrow 0_+} q(0, t). \quad (10)$$

The higher-order terms contain coefficients which are determined by the temporal partial derivatives $\partial_t^{(k)} q(0, 0_+)$. In what follows we introduce a high-order time-reconstruction procedure as a new way to compute the terms of the Taylor series expansion that involve time derivatives. The main idea is to exploit the history of the numerical solution to set up a reconstruction problem at the element interface. Several strategies may be proposed in this framework. Here, however, we discuss just two methods, namely, the TR0 and TR1 methods. Then, special emphasis is also given to the possibility of improving the reconstruction techniques via a non-linear procedure.

2.1 TR0 Method

As for the Toro–Titarev solver [29, 33, 37], to determine the leading term $q(0, 0_+)$ of the expansion (9) one solves the classical Riemann problem,

$$\left\{ \begin{array}{l} \text{PDEs : } \partial_t q + \partial_x f(q) = 0, \\ \text{ICs : } q(x, 0) = \begin{cases} q_L(0) \equiv \lim_{x \rightarrow 0_-} q_L(x) & \text{if } x < 0, \\ q_R(0) \equiv \lim_{x \rightarrow 0_+} q_R(x) & \text{if } x > 0, \end{cases} \end{array} \right. \quad (11)$$

where $q_L(0)$ and $q_R(0)$ are the boundary extrapolated values of the initial conditions in (8). As regards the determination of the higher-order terms of the expansion, we propose approximations to the time derivatives in (9) by means of a reconstructed function $v_{i+\frac{1}{2}}(t)$ located at the interface. More specifically, we assume that this function is a polynomial of degree K .

The TR0 solution method may be summarized as follows:

Step I *Leading term.* As previously claimed, the leading term is found by solving the classical homogeneous Riemann problem (11). Accordingly, we have

$$q(0, 0_+) = d^{(0)}(0) \quad (12)$$

with $d^{(0)}(\xi)$ denoting the self-similar solution of (11). We remark that the value of the leading term needs to be properly stored since it will be required by the solution technique at subsequent time levels.

Step II *Reconstruction polynomial.* To construct the polynomial $v_{i+\frac{1}{2}}(t)$ we consider the backward stencil of $K + 1$ time values

$$S_{K+1} = \{t^n, t^{n-1}, \dots, t^{n-K}\}, \quad (13)$$

and take advantage of the history of the solution. We thus select the set of leading terms

$$\tilde{q}_{K+1}^{(0)} = \{q(0, t_+^{n-k})\}_{k=0}^K, \quad (14)$$

obtained as in (12) and available from computations performed at previous time steps and at the current time level n (for $k = 0$). At this point, the reconstruction polynomial $v_{i+\frac{1}{2}}(t)$ is built as the unique polynomial of degree K that interpolates the set of leading terms $\tilde{q}_{K+1}^{(0)}$ on the stencil S_{K+1} , that is

$$v_{i+\frac{1}{2}}(t) = \{w(t) \in \mathcal{P}^K : w(t_+^{n-k}) = q(0, t_+^{n-k}), \forall t_+^{n-k} \in S_{K+1}\}. \quad (15)$$

Step III *Approximation of the time derivatives.* Once the reconstructed function $v_{i+\frac{1}{2}}(t)$ is available, its time derivatives evaluated at $t = t^n$ may be used to approximate the ones appearing in (9), namely

$$\partial_t^{(k)} q(0, 0_+) \approx \frac{d^k}{dt^k} v_{i+\frac{1}{2}}(t^n), \quad k = 1, \dots, K. \quad (16)$$

Step IV *The solution.* We can now form the approximate solution of the GRP_K at the interface position as the power series expansion,

$$q_{LR}(\tau) = C_0 + C_1 \tau + C_2 \tau^2 + \dots + C_K \tau^K \quad (17)$$

with

$$C_k \equiv \frac{v_{i+\frac{1}{2}}^{(k)}(t^n)}{k!}, \quad v_{i+\frac{1}{2}}^{(k)}(t^n) \equiv \frac{d^k}{dt^k} v_{i+\frac{1}{2}}(t^n), \quad k = 0, \dots, K. \quad (18)$$

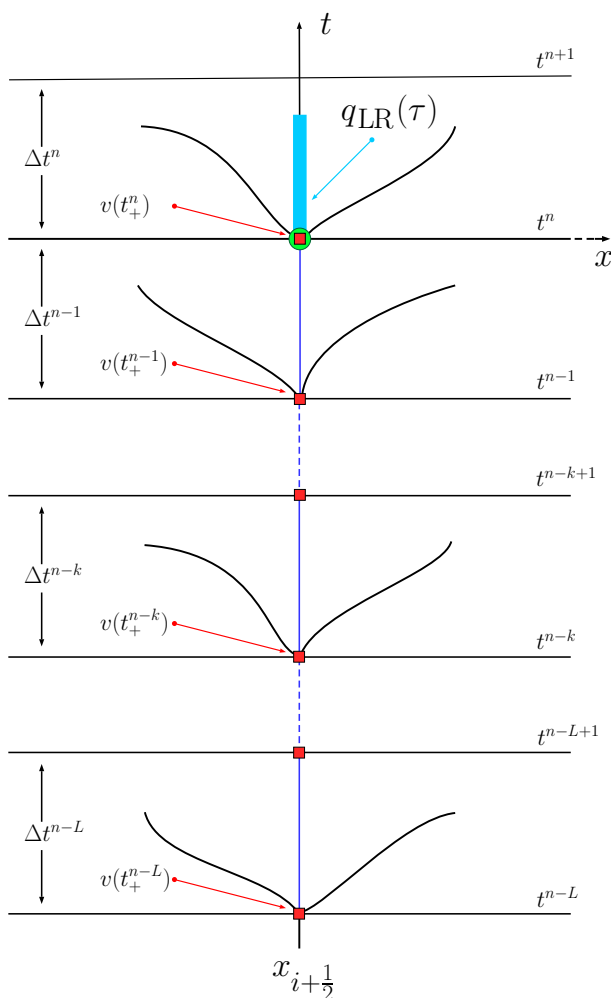
See Fig. 1 for a schematic representation of the TR0 method; take $L = K$ and $v_{i+\frac{1}{2}}^{(n-k)} = q(0, t_{i+\frac{1}{2}}^{n-k})$ for $k = 0, \dots, K$.

2.2 TR1 Method

For convenience we rewrite expression (9) as

$$q_{LR}(\tau) = q(0, 0_+) + \partial_t^{(1)} q(0, 0_+) \tau + \sum_{k=2}^K \left[\partial_t^{(k)} q(0, 0_+) \right] \frac{\tau^k}{k!}. \quad (19)$$

Fig. 1 Schematic representation of the time-reconstruction approach for the solution of the GRP_K. For the TR0 method take $L = K$ and $v_{i+\frac{1}{2}}^{(n-k)} = q(0, t_{i+\frac{1}{2}}^{n-k})$ for $k = 0, \dots, K$. Instead, for the TR1 method take $L = K - 1$ and $v_{i+\frac{1}{2}}^{(n-k)} = \partial_t^{(1)} q(0, t_{i+\frac{1}{2}}^{n-k})$ for $k = 0, \dots, K - 1$



In this way, we highlight that now both the leading term and the first-order term of the expansion are computed as for the Toro–Titarev solver [29, 33, 37]. Concerning higher-order terms (greater than or equal to two), we follow a time-reconstruction procedure very similar to the one presented above for the TR0 method. In particular, we propose to approximate the high-order time derivatives in (19) by means of a reconstructed function $v_{i+\frac{1}{2}}(t)$ located at the interface position. This reconstructed function is again assumed to be a polynomial, but of degree $K - 1$.

The TR1 solution method for the GRP_K includes the following steps:

Step I *Leading term and first-order term.* The leading term and the first-order term are obtained as for the Toro–Titarev solver. For details see [29, 33, 37]. Therefore, the leading term is determined as the similarity solution $d^{(0)}(0)$ of the classical Riemann problem (11), whereas the computation of the first-order term requires two distinct sub-steps. Following [37], we first solve the homogeneous linearized Riemann problem for the first-order spatial derivative

$$\begin{cases} \text{PDEs : } \partial_t(\partial_x^{(1)} q(x, t)) + f'(q(0, 0_+)) \partial_x(\partial_x^{(1)} q(x, t)) = 0, \\ \text{ICs : } \partial_x^{(1)} q(x, 0) = \begin{cases} q_L^{(1)}(0) \equiv \lim_{x \rightarrow 0_-} \frac{d}{dx} q_L(x) & \text{if } x < 0, \\ q_R^{(1)}(0) \equiv \lim_{x \rightarrow 0_+} \frac{d}{dx} q_R(x) & \text{if } x > 0, \end{cases} \end{cases} \quad (20)$$

whose self-similar solution is denoted as

$$\partial_x^{(1)} q(0, 0_+) = d^{(1)}(0). \quad (21)$$

Note that we are using the Cauchy–Kowalewskaya procedure to express the first-order time derivative in terms of the flux gradient and the source term (indeed very simple), namely

$$\partial_t q^{(1)}(0, 0_+) = -f'(q(0, 0_+)) \partial_x^{(1)} q(0, 0_+) + s(q(0, 0_+)). \quad (22)$$

The solution of (20) provides the spatial gradient to evaluate the first term of the right-hand side of (22). See [37] for details. Finally, we remark that the value of the first-order time derivative (22) needs to be stored, since it will be required by the solution technique at subsequent time levels.

Step II *Reconstruction polynomial.* In the case of the TR1 method, the reconstruction procedure takes place on the backward stencil of K time values

$$S_K = \{t^n, t^{n-1}, \dots, t^{n-K+1}\}. \quad (23)$$

Now, instead of considering the leading terms, we select the set of K first-order time derivatives

$$\bar{q}_K^{(1)} = \{\partial_t q^{(1)}(0, t_+^{n-k})\}_{k=0}^{K-1}. \quad (24)$$

We recall that these terms are obtained as explained in Step I and are available from computations performed at previous time steps or at the current time level n . Consequently, the reconstructed function $v_{i+\frac{1}{2}}(t)$ is defined as the unique polynomial of degree $K - 1$ that interpolates the set of first-order time derivatives $\bar{q}_K^{(1)}$ on the stencil S_K . Thus, we have

$$v_{i+\frac{1}{2}}(t) = \{ w(t) \in \mathcal{P}^{K-1} : w(t^{n-k}) = \partial_t^{(1)} q(0, t_+^{n-k}), \forall t^{n-k} \in S_K \}. \quad (25)$$

Step III *Approximation of the time derivatives.* As before, the evaluation at $t = t^n$ of the derivatives of $v_{i+\frac{1}{2}}(t)$ allows us to approximate the time derivatives appearing in (19). In this case we obtain

$$\partial_t^{(k)} q(0, 0_+) \approx \frac{d^{k-1}}{dt^{k-1}} v_{i+\frac{1}{2}}(t^n), \quad k = 2, \dots, K. \quad (26)$$

Step IV *The solution.* We can now form the approximate solution of the GRP_K at the interface position as the power series expansion

$$q_{LR}(\tau) = c_0 + c_1 \tau + c_2 \tau^2 + \dots + c_K \tau^K \quad (27)$$

with $c_0 \equiv q(0, 0_+)$ and

$$c_k \equiv \frac{v_{i+\frac{1}{2}}^{(k-1)}(t^n)}{k!}, \quad k = 1, \dots, K. \quad (28)$$

For a schematic representation of the TR1 method see again Fig. 1; in this case consider $L = K - 1$ and $v(t^{n-k}) = \partial_t^{(1)} q(0, t_+^{n-k})$ for $k = 0, \dots, K - 1$.

2.3 Efficient Implementation of the TR Approach

The most basic building block for the reconstruction techniques described above is certainly represented by the polynomial interpolation problem on a stencil of time values. As previously shown the task at hand is precisely the one to recover the time derivatives of the function $v_{i+\frac{1}{2}}(t)$ at the time level n . It is important to emphasise that, even though the idea behind this approach is quite straightforward, some care is required in its efficient implementation.

In this subsection we provide the necessary background for implementing the time-reconstruction algorithms in a very efficient way. In particular, we illustrate how to reduce the whole procedure to very simple and convenient matrix–vector multiplications. To this end, let us consider the temporal stencil of $L + 1$ time values

$$S_{L+1} = \{t^n, t^{n-1}, \dots, t^{n-L}\} \quad (29)$$

along with the set of known data

$$\bar{v}_{L+1} = \{v(0, t_+^{n-k})\}_{k=0}^L. \quad (30)$$

We recall that for the TR0 method we have that

$$\bar{v}_{L+1} \equiv \bar{q}_{K+1}^{(0)} \text{ and } L = K,$$

whereas for the TR1 method

$$\bar{v}_{L+1} \equiv \bar{q}_K^{(1)} \text{ and } L = K - 1.$$

We also remind the reader that the reconstructed function $v_{i+\frac{1}{2}}(t)$ is defined as the unique polynomial of degree L that interpolates the set of data $\bar{v}_{L+1}$ on the stencil S_{L+1} , namely

$$v_{i+\frac{1}{2}}(t) = \{ w(t) \in \mathcal{P}^L : w(t^{n-k}) = v(0, t_+^{n-k}), \forall t^{n-k} \in S_{L+1} \}. \quad (31)$$

In the reconstruction process the Lagrange form of the interpolation polynomial may be conveniently used,

$$v_{i+\frac{1}{2}}(t) \triangleq \sum_{k=0}^L v(t_+^{n-k}) L_k(t) \quad (32)$$

with L_k denoting the k th Lagrange polynomial defined as follows:

$$L_k(t) = \prod_{\substack{q=0 \\ q \neq k}}^L (t - t^{n-q}) \underbrace{\prod_{\substack{q=0 \\ q \neq k}}^L \frac{1}{t^{n-k} - t^{n-q}}}_{\triangleq \beta_k}. \quad (33)$$

Since we are interested in the time derivatives of $v_{i+\frac{1}{2}}(t)$ (up to the L th one), differentiating the polynomial m times, leads to

$$v_{i+\frac{1}{2}}^{(m)}(t) \equiv \frac{d^m}{dt^m} v_{i+\frac{1}{2}}(t) = \sum_{k=0}^L v(t_+^{n-k}) \frac{d^m}{dt^m} L_k(t), \quad m = 0, \dots, L. \quad (34)$$

After some algebraic manipulation, a general expression for the m th derivative of the Lagrange polynomial can be written as

$$\frac{d^m}{dt^m} L_k(t) = \beta_k m! \left(\sum_{\substack{I \in \{0, \dots, L\} \setminus \{k\} \\ \#I = L-m}} \prod_{l \in I} (t - t^{n-l}) \right), \quad m = 0, \dots, L. \quad (35)$$

We observe that in the trivial case of $m = 0$, the expression (35) correctly reduces to (33). Then, plugging into (34) results in

$$v_{i+\frac{1}{2}}^{(m)}(t) = \sum_{k=0}^L v(t_+^{n-k}) \left[\beta_k m! \left(\sum_{\substack{I \in \{0, \dots, L\} \setminus \{k\} \\ \#I = L-m}} \prod_{l \in I} (t - t^{n-l}) \right) \right]. \quad (36)$$

As we only need the derivatives of the reconstruction polynomial at the time level n , we can evaluate expression (36) at $t = t^n$ obtaining

$$v_{i+\frac{1}{2}}^{(m)}(t^n) = \sum_{k=0}^L v(t_+^{n-k}) \underbrace{\left[\beta_k m! \left(\sum_{\substack{I \in \{0, \dots, L\} \setminus \{k\} \\ \#I = L-m}} \prod_{l \in I} \left(\sum_{s=1}^l \Delta t^{n-s} \right) \right) \right]}_{\triangleq d_{m,k}}. \quad (37)$$

Now, by inspecting expression (37), one may observe that, as expected, the time-reconstruction problem is linear with respect to the set of interpolated values $\bar{v}_{L+1}$. As a consequence, it can be conveniently expressed as a matrix vector multiplication. We note, however, that in the case of non-uniform time steps, the matrix coefficients $d_{m,k}$ in general depend on the whole set of time step sizes Δt^{n-s} for $s = 1, \dots, L$. Another comment on expression (37) is that for $m = 0$ one correctly obtains $d_{0,k} = \delta_{0,k}$, with δ_{ij} representing the Kronecker delta. The sought time derivatives of the reconstruction polynomial at the time level n can therefore be written in compact form as

$$\begin{pmatrix} v_{i+\frac{1}{2}}^{(0)}(t^n) \\ v_{i+\frac{1}{2}}^{(1)}(t^n) \\ \vdots \\ v_{i+\frac{1}{2}}^{(L)}(t^n) \end{pmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & \dots & 0 \\ d_{1,0} & d_{1,1} & \dots & d_{1,L} \\ \vdots & \vdots & \ddots & \vdots \\ d_{L,0} & d_{L,1} & \dots & d_{L,L} \end{bmatrix}}_{\triangleq D_L} \underbrace{\begin{pmatrix} v_+^{(t^n)} \\ v_+^{(t^{n-1})} \\ \vdots \\ v_+^{(t^{n-L})} \end{pmatrix}}_{\triangleq \bar{v}_{L+1}}. \quad (38)$$

In the case of constant time steps the whole reconstruction procedure can be considerably simplified. First, we can simplify β_k as

$$\beta_k = \prod_{\substack{q=0 \\ q \neq k}}^L \frac{1}{(t^{n-k} - t^{n-q})} = \prod_{\substack{q=0 \\ q \neq k}}^L \frac{1}{(k-q)\Delta t} = \frac{(-1)^k}{k!(L-k)!(\Delta t)^L}. \quad (39)$$

Then, inserting (39) into the expression for $d_{m,k}$, yields

$$\begin{aligned} d_{m,k} &= \frac{(-1)^k m!}{k!(L-k)!(\Delta t)^L} \left(\sum_{\substack{l \in \{0, \dots, L\} \setminus \{k\} \\ \#l = L-m}} \prod_{l \in l} \left(\prod_{s=1}^l \Delta t^{n-s} \right) \right) \\ &= \frac{(-1)^k m!}{k!(L-k)!(\Delta t)^L} \left(\sum_{\substack{l \in \{0, \dots, L\} \setminus \{k\} \\ \#l = L-m}} \prod_{l \in l} (l\Delta t) \right) \\ &= \underbrace{\frac{1}{(\Delta t)^m} \frac{(-1)^k m!}{k!(L-k)!} \left(\sum_{\substack{l \in \{0, \dots, L\} \setminus \{k\} \\ \#l = L-m}} \prod_{l \in l} (l) \right)}_{\triangleq \tilde{d}_{m,k}}. \end{aligned} \quad (40)$$

Consequently, when the time step sizes are constant, (38) may be rewritten as

$$\begin{pmatrix} v_{i+\frac{1}{2}}^{(0)}(t^n) \\ v_{i+\frac{1}{2}}^{(1)}(t^n) \\ \vdots \\ v_{i+\frac{1}{2}}^{(L)}(t^n) \end{pmatrix} = \begin{bmatrix} 1 & & & \\ & \frac{1}{\Delta t} & & \\ & & \ddots & \\ & & & \frac{1}{(\Delta t)^L} \end{bmatrix} \underbrace{\begin{bmatrix} 1 & 0 & \dots & 0 \\ \tilde{d}_{1,0} & \tilde{d}_{1,1} & \dots & \tilde{d}_{1,L} \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{d}_{L,0} & \tilde{d}_{L,1} & \dots & \tilde{d}_{L,L} \end{bmatrix}}_{\triangleq \tilde{D}_L} \underbrace{\begin{pmatrix} v_+^{(t^n)} \\ v_+^{(t^{n-1})} \\ \vdots \\ v_+^{(t^{n-L})} \end{pmatrix}}_{\triangleq \bar{v}_{L+1}}. \quad (41)$$

We may note that now the coefficients $\tilde{d}_{m,k}$ do not depend on anything but m, k, L and can thus be computed beforehand, speeding up the process. For the sake of completeness, the coefficients of $\tilde{D}_L$ are given up to $L = 5$ as follows: $\tilde{D}_0 = [1]$, $\tilde{D}_1 = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$

$$\tilde{D}_2 = \begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & -2 & \frac{1}{2} \\ 1 & -2 & 1 \end{bmatrix}, \quad \tilde{D}_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{11}{6} & -3 & \frac{3}{2} & -\frac{1}{3} \\ 2 & -5 & 4 & -1 \\ 1 & -3 & 3 & -1 \end{bmatrix}, \quad \tilde{D}_4 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{25}{12} & -4 & 3 & -\frac{4}{3} & \frac{1}{4} \\ \frac{35}{12} & -\frac{26}{3} & \frac{19}{2} & -\frac{14}{3} & \frac{11}{12} \\ \frac{5}{2} & -9 & 12 & -7 & \frac{3}{2} \\ 1 & -4 & 6 & -4 & 1 \end{bmatrix},$$

$$\tilde{D}_5 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{137}{60} & -5 & 5 & -\frac{10}{3} & \frac{5}{4} & -\frac{1}{5} \\ \frac{15}{4} & -\frac{77}{6} & \frac{107}{6} & -13 & \frac{61}{12} & -\frac{5}{6} \\ \frac{17}{4} & -\frac{71}{4} & \frac{59}{2} & -\frac{49}{2} & \frac{41}{4} & -\frac{7}{4} \\ 3 & -14 & 26 & -24 & 11 & -2 \\ 1 & -5 & 10 & -10 & 5 & -1 \end{bmatrix}.$$

Using expression (38) or (41) we are therefore provided with all the necessary information to approximate the solution of the generalized Riemann problem at the interface. All the time derivatives appearing in (9) can be readily computed.

2.4 Non-linear Reconstruction

The TR0 and TR1 techniques, as presented above, involve a reconstruction procedure that has a linear nature (see (38) and (41)). The reconstructed function $v_{i+\frac{1}{2}}(t)$ at the interface is indeed a polynomial built on a fixed stencil. In this subsection we describe how to develop a non-linear reconstruction procedure that applies to both the TR0 and the TR1 methods (the resulting solvers are here called TR0W and TR1W, respectively). In this endeavour, we draw inspiration from a reconstruction technique, originally introduced by Shu and Tan in [25] to handle spatial reconstruction at the boundaries. The idea is to implement a time-reconstruction of WENO type, but on adaptive stencils of variable length.

Let us consider once again the standard stencil of $L + 1$ time values

$$S_{L+1} = \{t^n, t^{n-1}, \dots, t^{n-L}\} \quad (42)$$

along with the $L + 1$ candidate sub-stencils of different lengths which can be obtained from it, namely

$$s_{r+1} = \{t^n, t^{n-1}, \dots, t^{n-r}\}, \quad r = 0, \dots, L. \quad (43)$$

We wish to stress that all these sub-stencils contain the current time level n . Moreover, for the case $r = 0$ the sub-stencil includes only the time value t^n , while for $r = L$ the sub-stencil is coincident with S_{L+1} . We now propose to express the reconstructed function $v_{i+\frac{1}{2}}(t)$ as a convex combination of polynomials of different degrees, that is

$$v_{i+\frac{1}{2}}(t) = \sum_{r=0}^L \omega_r \tilde{v}_r(t), \quad (44)$$

where ω_r are non-linear weights computed as will be shown later and such that $\sum_{r=0}^L \omega_r = 1$. As regards the functions $\tilde{v}_r(t)$, they are defined as the polynomials of degree r that interpolate the set of data

$$\tilde{v}_{r+1} = \left\{ v(0, t_+^{n-k}) \right\}_{k=0}^r \quad (45)$$

on the sub-stencil s_{r+1} . Thus, we have

$$\tilde{v}_r(t) = \{ w(t) \in \mathcal{P}^r : w(t_+^{n-k}) = v(0, t_+^{n-k}), \forall t_+^{n-k} \in s_{r+1} \}. \quad (46)$$

For convenience we dropped the sub-index $i + \frac{1}{2}$ and gave emphasis to the degree of the polynomial. We recall, however, that even these functions are located at the interface. As it is clear from expression (44), the sought time derivatives of $v_{i+\frac{1}{2}}(t)$ at $t = t^n$ are now obtained as a weighted combination of those of the polynomials $\tilde{v}_r(t)$. We remark that the derivatives of $\tilde{v}_r(t)$ at $t = t^n$ may in turn be computed in an entirely analogous manner to the case presented in Sect. 2.3, using r instead of L .

Clearly, in the development of such a procedure, the determination of the non-linear weights plays a primary role and requires special care. Following the guidelines of [26], we make some useful observations. First of all, when the set of data $\tilde{v}_{L+1}$ is smooth on the main stencil S_{L+1} , we want that the polynomial of the highest degree, i.e., L , dominates the convex combination (44); this in fact guarantees that the designed order of accuracy is preserved on smooth data. In this case, we therefore would like to have

$$\begin{cases} \omega_r = (\Delta t)^{L-r}, & r = 0, \dots, L-1; \\ \omega_L = 1 - \sum_{r=0}^{L-1} \omega_r \end{cases} \quad (47)$$

with Δt denoting, for simplicity, the time step size at the time level n . These weights are called prototype weights and allow to maintain the desired order of accuracy on smooth data, as claimed; from now on we denote them by γ_r . On the other hand, when a discontinuity is present inside S_{L+1} , a possibly lower accurate but more robust procedure is preferable. In this situation we wish that only the lower-degree polynomials on sub-stencils with smooth data provide a significant contribution in (44). This is accomplished by defining the non-linear weights as follows:

$$\begin{cases} \omega_r = \frac{\hat{\omega}_r}{\sum_{r=0}^L \hat{\omega}_r}, \\ \hat{\omega}_r = \frac{\gamma_r}{(\epsilon + \beta_r)^q} \end{cases} \quad (48)$$

with $q = 2$, $\epsilon = 10^{-6}$ and β_r the improved smoothness indicators originally introduced by Jiang and Shu in [12] and defined, for the time-reconstruction, as

$$\beta_r = \sum_{l=1}^r \int_{t^n}^{t^n + \Delta t} (\Delta t)^{2l-1} \left(\frac{d^l}{dt^l} \tilde{v}_r(t) \right)^2 dt, \quad r = 1, \dots, L. \quad (49)$$

By carefully examining (49), we observe that the smoothness indicators are well defined for $r > 0$, while for the constant polynomial the indicator is identically zero, rendering it unusable to gauge the smoothness of the polynomial. To overcome this limitation we follow the work of Kall in [13] and choose

$$\beta_0 = \beta_1 \left(\frac{\beta_1}{\beta_2} \right)^2, \quad (50)$$

assuming that β_1 and β_2 are available. For smooth data this choice assigns to β_0 roughly the same value as the other smoothness indicators have, which in turn makes the higher order polynomials on the longer sub-stencils dominate the convex combination (44). On the other hand, when there is a discontinuity within s_2 or s_3 , the ratio $\frac{\beta_1}{\beta_2}$ is much smaller than one. Hence, via (50), β_0 is close to zero, making the constant reconstruction polynomial dominant in (44). Furthermore, since the smoothness indicators (49) scale quadratically with the data, this gives an approximately scaling invariant reconstruction procedure. At this stage, it is worth remarking that the assumption $\beta_2 \gg \beta_1$ for discontinuous data is usually, but not always, correct. To date, however, we have not encountered any problems when using this technique in numerical simulations.

Concerning the computations of the smoothness indicators (49), one notes that they may be written as positive semi-definite quadratic forms of the underlying data. The complex expression (49) thus reduces to

$$\beta_r = \bar{v}_{r+1}^T \mathcal{B}_r \bar{v}_{r+1}, \quad r = 1, \dots, L \quad (51)$$

with $\mathcal{B}_r \in \mathbb{R}^{(r+1) \times (r+1)}$ the symmetric positive semi-definite matrices related to the quadratic forms. The determination of the coefficients of $\mathcal{B}_r$ is quite cumbersome and can be performed with the help of a symbolic manipulator. Moreover, for the general case of non-uniform time steps, the coefficients depend on the whole set of time step sizes $\{\Delta t^{n-k}\}_{k=0}^r$. On the other hand, in the particular case in which the time steps are constant the matrices can be precomputed and stored, which significantly speeds up the computation of the smoothness indicators. For the sake of completeness, the coefficients of $\mathcal{B}_r$ for uniform time step sizes are given up to $r = 5$ as follows: $\mathcal{B}_1 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$,

$$\mathcal{B}_2 = \begin{bmatrix} \frac{61}{12} & -\frac{49}{6} & \frac{37}{12} \\ -\frac{49}{6} & \frac{40}{3} & -\frac{31}{6} \\ \frac{37}{12} & -\frac{31}{6} & \frac{25}{12} \end{bmatrix}, \quad \mathcal{B}_3 = \begin{bmatrix} \frac{1517}{90} & -\frac{4663}{120} & \frac{603}{20} & -\frac{2933}{360} \\ -\frac{4663}{120} & \frac{1373}{15} & -\frac{8699}{120} & \frac{1189}{60} \\ \frac{603}{20} & -\frac{8699}{120} & \frac{878}{15} & -\frac{1943}{120} \\ -\frac{2933}{360} & \frac{1189}{60} & -\frac{1943}{120} & \frac{407}{90} \end{bmatrix},$$

$$\mathcal{B}_4 = \begin{bmatrix} \frac{2123}{45} & -\frac{4641}{32} & \frac{20381}{118} & -\frac{4943}{52} & \frac{1797}{89} \\ -\frac{4641}{32} & \frac{9571}{21} & -\frac{7177}{13} & \frac{61092}{199} & -\frac{11751}{179} \\ \frac{20381}{118} & -\frac{7177}{13} & \frac{45411}{67} & -\frac{35739}{94} & \frac{16193}{198} \\ -\frac{4943}{52} & \frac{61092}{199} & -\frac{35739}{94} & \frac{22109}{103} & -\frac{3850}{83} \\ \frac{1797}{89} & -\frac{11751}{179} & \frac{16193}{198} & -\frac{3850}{83} & \frac{2525}{251} \end{bmatrix},$$

$$\mathcal{B}_5 = \begin{bmatrix} \frac{8531}{70} & -\frac{20446}{43} & \frac{9251}{12} & -\frac{34269}{53} & \frac{15291}{55} & -\frac{11111}{228} \\ -\frac{20446}{43} & \frac{327488}{173} & -\frac{65372}{21} & \frac{50056}{19} & -\frac{46729}{41} & \frac{17858}{89} \\ \frac{9251}{12} & -\frac{65372}{21} & \frac{196549}{38} & -\frac{52897}{12} & \frac{26829}{14} & -\frac{17945}{53} \\ -\frac{34269}{53} & \frac{50056}{19} & -\frac{52897}{12} & \frac{71741}{19} & -\frac{34600}{21} & \frac{76774}{263} \\ \frac{15291}{55} & -\frac{46729}{41} & \frac{26829}{14} & -\frac{34600}{21} & \frac{124732}{173} & -\frac{5889}{46} \\ -\frac{11111}{228} & \frac{17858}{89} & -\frac{17945}{53} & \frac{76774}{263} & -\frac{5889}{46} & \frac{9063}{398} \end{bmatrix}.$$

Once the smoothness indicators are available, the non-linear weights can be computed as in (48).

For a clearer understanding of the advantages of a non-linear reconstruction procedure, consider the example illustrated in Fig. 2. In the depicted situation the set of interpolated data originates from a discontinuous function. Since the discontinuity is located inside the interval $[t^{n-3}, t^{n-2}]$, only the data on the sub-stencils s_1, s_2 and s_3 are smooth. In this situation a standard linear procedure would take into account only the quartic polynomial on the main stencil S_5 . As a consequence, since the data are non-smooth on S_5 , the reconstructed function (blue circles) on the time interval $[t^n, t^{n+1}]$ would dramatically fail to capture the behaviour of the underlying discontinuous function. Conversely, a non-linear reconstruction procedure weights all the polynomials (from the zero degree up to the fourth one) on the basis of the smoothness of the data. More precisely, in the depicted situation the convex combination is dominated by the quadratic polynomial, which is the one of highest degree amongst those constructed on smooth data. As we can clearly see, a non-linear reconstruction is highly suggested in this situation.

We can therefore state that the utilisation of a non-linear reconstruction procedure for the solution of the GRP_K is necessary to improve the robustness of the time-reconstruction solvers in the presence of discontinuities.

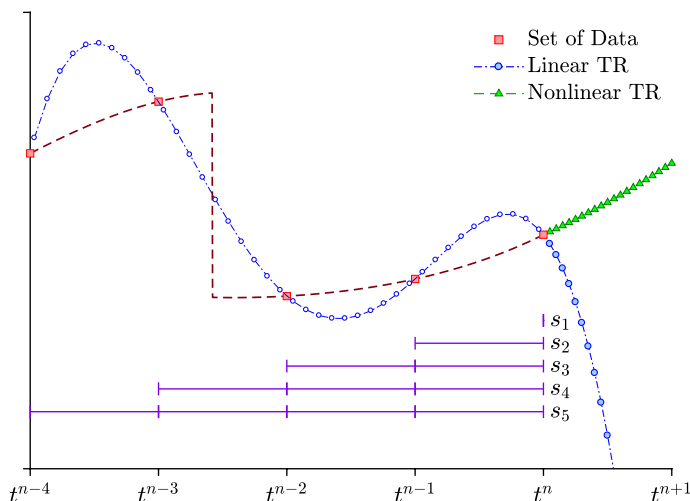


Fig. 2 Comparison between the linear and non-linear time-reconstruction procedures when applied to non-smooth data. The linear procedure completely fails to capture the behaviour of the underlying discontinuous function in the time interval $[t^n, t^{n+1}]$

3 ADER TR Schemes

The GRP_K solvers described in the previous section can be used to construct ADER methods of arbitrary order of accuracy in both space and time. In the following, such schemes are called ADER-TR to distinguish them from conventional ADER methods. In this section we first deal with the construction of these simplified ADER schemes and then we investigate thoroughly their linear stability properties. It is worth remarking that ADER-TR schemes are simplified in the sense that the complexity of the Cauchy–Kowalewskaya procedure is avoided.

3.1 The Numerical Scheme

Here we introduce the main features of ADER-TR schemes as applied to the scalar conservation law with a source term (2). We recall that ADER schemes are one-step, fully-discrete and of arbitrary order of accuracy in both space and time. In a finite-volume framework, they are based on the explicit conservative formula,

$$q_i^{n+1} = q_i^n - \frac{\Delta t}{\Delta x} \left(f_{i+\frac{1}{2}} - f_{i-\frac{1}{2}} \right) + \Delta t s_i. \quad (52)$$

As usual, what is crucial in a finite-volume method is the design of a strategy to compute the numerical flux $f_{i+\frac{1}{2}}$ and the numerical source s_i appearing in (52). In what follows we show how to obtain these key ingredients for the case of ADER-TR schemes by deploying the time reconstruction approach addressed in the previous section. Before proceeding, however, we wish to clarify that the present schemes generate high-order approximations by following the one-step formula (52). The reconstruction in time only avoids the use of the Cauchy–Kowalewskaya procedure and the marching in time is still achieved by the ADER approach.

3.1.1 The Numerical Flux

The numerical flux $f_{i+\frac{1}{2}}$ results from approximating the time-integral average ($\hat{f}_{i+\frac{1}{2}}$ in (4) of the physical flux function at the cell interface. For convenience we introduce local coordinates in which x is replaced by $x - x_{i+\frac{1}{2}}$ and t is replaced by $t - t^n$. In the ADER approach, the numerical flux is computed by evaluating the integral

$$f_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_0^{\Delta t} f(q_{\text{LR}}(\tau)) d\tau, \quad (53)$$

with a suitably high-order numerical integration scheme, where $q_{\text{LR}}(\tau)$ is a function of time and represents the solution of a local generalized Riemann problem. As originally proposed by Toro et al. [35], the initial states for this local generalized Riemann problem are the reconstruction polynomials $p_i(x)$ and $p_{i+1}(x)$ defined on the two consecutive cells i and $i+1$ and obtained from a high-order conservative and non-oscillatory reconstruction procedure, typically a WENO procedure [12, 18]. In this regard, it is important to remark that for a J th order accurate scheme the reconstruction polynomials are of degree $J-1$.

To construct ADER-TR schemes, the solution of the generalized Riemann problem $q_{\text{LR}}(\tau)$ at the cell interface may be conveniently split into a low-order part $q_{\text{LR}}^{(l)}(\tau)$ and a high-order part $q_{\text{LR}}^{(h)}(\tau)$, namely

$$q_{\text{LR}}(\tau) = q_{\text{LR}}^{(l)}(\tau) + q_{\text{LR}}^{(h)}(\tau). \quad (54)$$

While the terms in $q_{\text{LR}}^{(l)}(\tau)$ are obtained in a conventional manner, as for the Toro–Titarev solver [29, 33, 37], those in $q_{\text{LR}}^{(h)}(\tau)$ are computed from a time-reconstruction polynomial $v_{i+\frac{1}{2}}(t)$ whose coefficients are in turn evaluated using the history of the numerical solution, as explained in detail in Sect. 2. For example, in the case of a J th order ADER-TR1 scheme the two contributions read

$$\begin{cases} q_{\text{LR}}^{(l)}(\tau) = q(0, 0_+) + [-f'(q(0, 0_+))\partial_x^{(1)}q(0, 0_+) + s(q(0, 0_+))]\tau, \\ q_{\text{LR}}^{(h)}(\tau) = \sum_{k=2}^{J-1} \left[v_{i+\frac{1}{2}}^{(k-1)}(t^n) \right] \frac{\tau^k}{k!}. \end{cases} \quad (55)$$

Note, however, that in the case of the TR0 method, the low-order part of (54) contains only the leading term $q(0, 0_+)$ displayed in the first line of (55), while the second term in the same line becomes part of the high-order component. It is worth mentioning that one can also consider to increase the number of terms in $q_{\text{LR}}^{(l)}(\tau)$, but this is not pursued here.

At this stage, a relevant observation is that it is the high-order part $q_{\text{LR}}^{(h)}(\tau)$ that requires information from the history of the numerical solution and hence confers a multi-level character to the ADER-TR methods presented here. Accordingly, ADER-TR schemes may be regarded as one-step multi-level methods; due to this fact, some additional difficulties arise naturally. First, ADER-TR schemes require more storage than conventional ADER methods, a disadvantage that is relevant primarily in multidimensional calculations. Furthermore, ADER-TR schemes require special start-up procedures, since initially only one level of data is known. To this end, the most natural choice is to use a conventional ADER scheme as a means of generating all the necessary starting values. However, we have also verified that the utilisation of a WENO scheme [1, 11, 12] on a suitably refined mesh represents a valid and simple alternative for the start-up process. This also allows to construct a numerical scheme that completely avoids the cumbersome Cauchy–Kowalewskaya procedure.

We finalise the discussion by recalling that an appropriate Gaussian quadrature rule is always necessary to numerically evaluate expression (53) to the desired accuracy. Therefore, the numerical flux is written as

$$f_{i+\frac{1}{2}} = \sum_{\alpha=1}^{K_\alpha} \omega_\alpha f(q_{\text{LR}}(\gamma_\alpha \Delta t)), \quad (56)$$

where γ_α and ω_α are properly scaled nodes and weights of the rule, respectively, and K_α is the number of nodes.

3.1.2 The Numerical Source

In the presence of source terms, the ADER approach determines the numerical source s_i by approximating the space–time integral $\hat{s}_i$ in (4) within the fixed control space–time volume $V = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \times [0, \Delta t]$ to obtain the average

$$s_i = \frac{1}{\Delta t} \frac{1}{\Delta x} \int_0^{\Delta t} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} s(q_i(x, t)) dx dt, \quad (57)$$

where $q_i(x, t)$ denotes an approximation of the solution $q(x, t)$ of (2) inside the space–time control volume V . In particular, at the local time $t = 0$ we have that $q_i(x, t)$ is coincident with the spatial reconstruction polynomial, namely

$$q_i(x, 0) = p_i(x). \quad (58)$$

A possible way to evaluate (57), retaining the desired order of accuracy J , is to consider a number of fixed integration points $\hat{x}_\alpha$ within the cell i , at which one can write a time power series expansion of the form

$$q_{\hat{x}_\alpha}(\tau) = q_i(\hat{x}_\alpha, 0_+) + \partial_t^{(1)} q_i(\hat{x}_\alpha, 0_+) \tau + \sum_{k=2}^{J-1} \partial_t^{(k)} q_i(\hat{x}_\alpha, 0_+) \frac{\tau^k}{k!}. \quad (59)$$

Using an appropriate Gaussian quadrature rule the numerical source may be written as

$$s_i = \sum_{\alpha=1}^{K_\alpha} \sum_{\beta=1}^{K_\beta} \omega_\alpha \omega_\beta s(q_{\hat{x}_\alpha}(\gamma_\beta \Delta t)), \quad (60)$$

where $\hat{x}_\alpha = x_{i-\frac{1}{2}} + \gamma_\alpha \Delta x$ and γ and ω are properly scaled nodes and weights of the rule, respectively.

Again, expansion (59) can be conveniently split into a low-order part, which is obtained in a standard manner, and a high-order part, which is computed using the time-reconstruction approach

$$q_{\hat{x}_\alpha}(\tau) = q_{\hat{x}_\alpha}^{(l)}(\tau) + q_{\hat{x}_\alpha}^{(h)}(\tau). \quad (61)$$

Although at this stage the procedure is analogous to that for the flux evaluation, some clarifications are in order:

- i) since $p_i(x)$ is continuous inside the cell i , the leading term of expansion (59) is obtained from point values of the reconstruction polynomial, i.e., $q_i(\hat{x}_\alpha, 0_+) = p_i(\hat{x}_\alpha)$. Note that in this case there is no need to solve any Riemann problem;
- ii) in the case of the TR1 method the first-order term of (59), which is included in $q_{\hat{x}_\alpha}^{(l)}(\tau)$, reads

$$\partial_t^{(1)} q_i(\hat{x}_\alpha, 0_+) = -f'(q_i(\hat{x}_\alpha, 0_+)) \frac{d}{dx} p_i(\hat{x}_\alpha) + s(q_i(\hat{x}_\alpha, 0_+)), \quad (62)$$

where the relation $\partial_x^{(1)} q_i(\hat{x}_\alpha, 0_+) = \frac{d}{dx} p_i(\hat{x}_\alpha)$ has been used;

- iii) all the time derivatives included in the high-order part $q_{\hat{x}_\alpha}^{(h)}(\tau)$ of the time Taylor series expansion are obtained from a reconstruction function $v_{\hat{x}_\alpha}(t)$ located at $x = \hat{x}_\alpha$, in accordance with the time-reconstruction approach described in Sect. 2. This means that the function $v_{\hat{x}_\alpha}(t)$ is a polynomial of appropriate degree which interpolates on a backward stencil of time values the set of leading terms of (59) (for the TR0 method) or the set of first-order time derivatives appearing in the expansion (59) (for the TR1 method). It is worth mentioning that the possibility to set up a non-linear time-reconstruction extends naturally to this context.

Finally, we remark that also for the evaluation of the numerical source s_j , a start-up algorithm needs to be used to provide all the temporal information required by the computation of the high-order part of (61).

3.2 Stability

A very important issue is the stability of the numerical methods just described. As pointed out by Toro and Titarev themselves in [30], conventional ADER schemes are stable, in practical calculations, under the usual Courant number limitations for both smooth and discontinuous solutions. Compared to standard ADER methods, simplified ADER-TR schemes have a main disadvantage: they have a reduced stability region, which in general decreases as the order of accuracy is increased. The main purpose of this section is therefore to investigate in more detail the linear stability properties of ADER-TR methods.

As is usual when performing the linear stability analysis, we consider the model linear one-dimensional advection equation

$$\partial_t q + \partial_x f(q) = 0, \quad f = \lambda q \quad (63)$$

with periodic boundary conditions. $\lambda > 0$ is the constant propagation speed. The stability of linear ADER-TR schemes is analysed by means of the von Neumann method. Due to the complexity of the resulting expressions, we adopt the idea of Colella [4] and Toro and Billet [34] to verify the stability condition numerically rather than analytically. For non-linear schemes, on the other hand, we deduce the stability regions from numerical experiments.

3.2.1 Von Neumann Analysis

We perform the von Neumann stability analysis for the linear version of ADER-TR methods. With this intention, we only consider schemes based on linear TR solvers. This is, however, not sufficient to construct linear methods. One must recall that, typically, ADER schemes also use the accurate non-linear weighted essentially non-oscillatory (WENO) reconstruction [12, 18] to locally represent the spatial variation of the solution within each cell. In this procedure the non-linearity is introduced by the weights which control the convex combination. One possible way of obtaining linear ADER schemes is to use a fixed stencil reconstruction. In this regard, we only take into account centred (for odd orders of accuracy) and upwind-biased (for even orders of accuracy) stencils.

It is worth recalling at this stage that ADER-TR schemes are multi-level, involving L previous time steps. For schemes implementing a TR0 solver we have $L = J - 1$, whereas for schemes with a TR1 solver we have $L = J - 2$, where J is the order of accuracy of the scheme. A compact expression of these linear methods can be given in the form

$$q_i^{n+1} = \sum_{l=0}^L \sum_p a_{p,l} q_{i+p}^{n-l}, \quad (64)$$

where $p = \{-\lfloor \frac{J}{2} \rfloor, \dots, \lceil \frac{J}{2} \rceil - 1\}$ and $a_{p,l}$ are constant coefficients. $\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$ denote the floor and the ceiling functions, respectively.

For linear ADER-TR schemes the assessment of stability properties is more complicated than in the case of one-step schemes, since the definition of the standard amplification factor has to be more general. Let us introduce

$$\mathbf{X}_i^n = \begin{pmatrix} q_i^n \\ q_i^{n-1} \\ \vdots \\ q_i^{n-L} \end{pmatrix}. \quad (65)$$

Then, the von Neumann stability analysis of the linear versions of our schemes is performed as follows: we consider a trial solution composed by an arbitrary harmonic function which has the form

$$\mathbf{X}_i^n = \mathbf{A}^n e^{I i \alpha}, \quad (66)$$

where $I = \sqrt{-1}$ and $\mathbf{A}$ is the complex amplitude vector of the generic harmonic with phase angle $\alpha = k\Delta x$ and wave number k . Inserting this expression inside (64), along with the trivial relation between the components of the vectors $\mathbf{A}^n$ and $\mathbf{A}^{n+1}$, that is

$$a_{l+1}^{n+1} = a_l^n, \quad l = 1, \dots, L, \quad (67)$$

we recover the standard one-step form

$$\mathbf{A}^{n+1} = \mathbf{G}(\alpha, c) \mathbf{A}^n. \quad (68)$$

Here $\mathbf{G}(\alpha, c)$ is the amplification matrix, an $(L+1) \times (L+1)$ function of the phase angle α and of the Courant number c . Following the guidelines of Trefethen in [39, 40], the stability condition can be expressed by the requirement

$$\|(\mathbf{G}(\alpha, c))^r\|_2 \leq C, \quad \forall \alpha \in (-\pi, \pi), \quad r \geq 1, \quad (69)$$

where, for clarity, we point out that $(\mathbf{G}(\alpha, c))^r$ is the r th power of the amplification matrix.

Stability is thus a question of power-boundedness of a family of $(L+1) \times (L+1)$ matrices, a family characterized by the two parameters α and c . The simplest estimates of $\|(\mathbf{G}(\alpha, c))^r\|$ are based on the norm $\|\mathbf{G}(\alpha, c)\|$ or the spectral radius $\rho(\mathbf{G}(\alpha, c))$, that is, the largest of the moduli of the eigenvalues of $\mathbf{G}(\alpha, c)$, namely

$$\rho(\mathbf{G}(\alpha, c)) = \max_{\lambda \in \Lambda(\mathbf{G}(\alpha, c))} |\lambda| \quad (70)$$

with $\Lambda(\mathbf{G}(\alpha, c))$ denoting the spectrum of the amplification matrix. These two quantities provide a lower and an upper bound on the stability condition (69) according to the easily proved inequalities

$$\rho(\mathbf{G}(\alpha, c))^r \leq \|\mathbf{G}(\alpha, c)^r\| \leq \|\mathbf{G}(\alpha, c)\|^r. \quad (71)$$

Then, it follows that:

- i) $\rho(\mathbf{G}(\alpha, c)) \leq 1$ is necessary for stability;
- ii) $\|\mathbf{G}(\alpha, c)\| \leq 1$ is sufficient for stability.

Condition i) is often referred to as the von Neumann condition for multi-level schemes. It is a statement about eigenvalues of amplification matrices and is necessary but not sufficient for stability. To eliminate the gap between conditions i) and ii) one can apply the Kreiss matrix theorem [14]. See the works of Spijker et al. [23], Reddy and Trefethen [22] and references therein for more details.

Let us introduce the ϵ -pseudospectral radius: for any matrix $\mathbf{B}$ and constant $\epsilon \geq 0$ we define the ϵ -pseudospectral radius $\rho_\epsilon(\mathbf{B})$ of $\mathbf{B}$ to be the largest of the moduli of its ϵ -pseudo eigenvalues, that is

$$\rho_\epsilon(\mathbf{B}) = \sup_{\lambda_\epsilon \in \Lambda_\epsilon(\mathbf{B})} |\lambda_\epsilon|, \quad (72)$$

where $\Lambda_\epsilon(\mathbf{B})$ is the ϵ -pseudospectrum of $\mathbf{B}$ defined as

$$\Lambda_\epsilon(\mathbf{B}) = \{\lambda \in \Lambda(\mathbf{B} + \mathbf{E}) : \|\mathbf{E}\|_2 \leq \epsilon\}. \quad (73)$$

The Kreiss matrix theorem can be restated in terms of the ϵ -pseudospectral radius as follows: a matrix $\mathbf{B}$ is power-bounded if and only if

$$\rho_\epsilon(\mathbf{B}) \leq 1 + \mathcal{O}(\epsilon), \quad \epsilon \rightarrow 0. \quad (74)$$

Consequently, the stability requirement (69) is equivalent to

$$\rho_\epsilon(\mathbf{G}(\alpha, c)) \leq 1 + \mathcal{O}(\epsilon), \quad \forall \alpha \in (-\pi, \pi), \quad \epsilon \rightarrow 0. \quad (75)$$

Condition (75) gives a powerful tool to estimate the stability region of our schemes. However, for linear ADER-TR schemes the expression of the amplification matrix $\mathbf{G}(\alpha, c)$ is rather complicated and is intractable for algebraic analysis. To overcome this difficulty, we adopt the idea of Colella [4] and Toro and Billet [34] to verify the stability condition numerically rather than analytically. In this case, drawing a contour plot of the ϵ -pseudospectral radius for a large number of values (c, α) in a rectangle $[c_{\min}, c_{\max}] \times [-\pi, \pi]$ gives a useful indication as to the linear stability region of the scheme; here $c_{\min}$ and $c_{\max}$ are chosen to be, respectively, smaller and larger than the expected stability limit. Condition (75) is in this way checked at a discrete level using the algorithm proposed by Mengi and Overton [19] to compute the ϵ -pseudospectral radius.

For illustration purposes, Fig. 3 depicts the contour plots of the ϵ -pseudospectral radius for the linear ADER-TR1 scheme of fifth-order of accuracy. We remark that, due to symmetric considerations, in the plot we only take into account values of α in the range $[0, \pi]$. The main results of the Von Neumann analysis are then summarized in Table 1, where we list the stability limits obtained for schemes up to the sixth order of accuracy.

Finally, for the sake of clarity, we report below the expression of the amplification matrix for the third-order linear ADER-TR1 scheme:

$$\mathbf{G}_3(\alpha, c) = \begin{bmatrix} g_1(\alpha, c) & g_2(\alpha, c) \\ 1 & 0 \end{bmatrix}$$

with $g_1(\alpha, c)$ and $g_2(\alpha, c)$ defined as

$$\begin{aligned} g_1(\alpha, c) &= 1 - \frac{c}{6}(e^{I\alpha} - 1)e^{-2I\alpha}[-w_1 - w_2 + (5e^{I\alpha} + 2e^{2I\alpha})(w_1 + w_2) \\ &\quad + 3c(e^{I\alpha} - e^{2I\alpha})(w_1q_1^2 + w_2q_2^2 + 2w_1q_1 + 2w_2q_2)]; \\ g_2(\alpha, c) &= 2c^2(w_1\gamma_1^2 + w_2\gamma_2^2)\sin^2\left(\frac{\alpha}{2}\right), \end{aligned}$$

Table 1 Linear stability limits for ADER-TR0 and ADER-TR1 methods. The stability limit of the second-order ADER-TR1 scheme is not considered, since it boils down to the second-order ADER-TT scheme

J	2	3	4	5	6
ADER-TR0	0.250	0.400	0.112	0.123	0.043
ADER-TR1	–	0.600	0.476	0.372	0.286

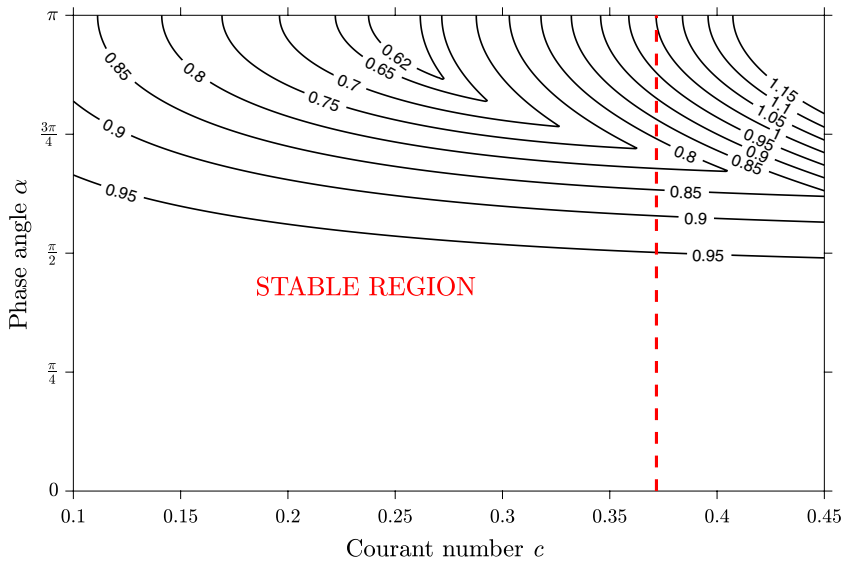


Fig. 3 Contour plot of the ϵ -pseudospectral radius for the fifth-order linear ADER-TR1 scheme. The stability region is detected by checking condition (75) at a discrete level. The stability limit is found as the first value of the Courant number that intersects the isoline $\rho_\epsilon = 1 + \mathcal{O}(\epsilon)$

where γ_1, γ_2 and w_1, w_2 are properly scaled abscissas and weights of the Gauss–Legendre quadrature rule. One notes that the expression is quite complex, even if the order of accuracy is rather low.

3.2.2 Empirical Stability Study

We recall that the von Neumann analysis we performed above is strictly valid only under the assumption of linear methods. In this subsection we thus present an empirical study aimed at determining the stability conditions for their non-linear counterparts. In the assessment of stability properties a very effective and simple tool is represented by numerical experiments. In this context, we conduct an exhaustive numerical testing, solving the linear advection equation (63) with a variety of initial conditions. Specifically, we consider the smooth but difficult initial data like

$$q(x, 0) = q_0(x) = \sin^4(\pi x) \quad (76)$$

and the quite sharply peaked Gaussian defined as [12]

$$q_0(x) = \frac{1}{6} [G(x, \gamma, z - \delta) + G(x, \gamma, z + \delta) + 4G(x, \gamma, z)], \quad (77)$$

where $G(x, \gamma, z) = e^{-\gamma(x-z)^2}$ with $\delta = 0.01$, $z = 0$ and $\gamma = \frac{\log 2}{369^2}$. In our simulations the numerical solutions are updated up to 10^4 time steps with the CFL coefficient left free to vary; the schemes are run on a periodic domain $[-1, 1]$ discretized by a uniform mesh of 100 cells.

The primary goal of the numerical experiments is to outline the CFL-type stability restrictions on the time step which guarantee, in the simple case of linear advection, that the numerical solution computed from a smooth initial profile remains smooth. In other words, we require that the numerical solution does not exhibit unstable oscillations whose

amplitude increases over time. To develop an empirical procedure, which aims to automatically provide the sought stability limits, we have implemented a bisection-like algorithm based on a logical index which mutates its value depending on whether the scheme is stable or not. A leading role inside this procedure is played by a smoothness indicator, which is capable of identifying the appearance of destabilizing oscillations within the numerical solution. To handle oscillations, a measure of smoothness is in fact necessary; the most commonly used measures for smoothness are the improved indicators introduced by Jiang and Shu [12]. For a sufficiently smooth differentiable function f the smoothness indicator of order k on cell I_i is defined as

$$IS_{I_i}^k(f) = \sum_{j=1}^{k-1} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \Delta x^{2j-1} \left(\frac{\partial^j f(x)}{\partial x^j} \right)^2 dx. \quad (78)$$

In particular, when the function f is the spatial reconstruction polynomial on the r shifted stencil $S_i^{k,r} = \{I_{i-r}, \dots, I_{i-r+k-1}\}$, the smoothness indicator is denoted as $IS_{I_i}^{k,r}$.

To detect, at the generic time level n , the appearance of oscillations, we compute for each cell the value of the centred (for odd orders of accuracy) or upwind-biased (for even orders of accuracy) smoothness indicator $IS_{I_i}^{J, \lfloor \frac{J-1}{2} \rfloor}$; here again J stands for the order of accuracy of the scheme and $\lfloor \cdot \rfloor$ denotes the floor function.

Simple but relevant pseudo-statistical quantities can be introduced to help in the study,

$$\left\{ \begin{array}{l} \mu_{IS} = \frac{1}{N} \sum_{i=1}^N IS_{I_i}^{J, \lfloor \frac{J-1}{2} \rfloor}, \\ \sigma_{IS} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N \left(IS_{I_i}^{J, \lfloor \frac{J-1}{2} \rfloor} - \mu_{IS} \right)^2}, \\ \delta_{IS} = \max_{I_i} \frac{|IS_{I_i}^{J, \lfloor \frac{J-1}{2} \rfloor} - \mu_{IS}|}{\sigma_{IS}}, \end{array} \right. \quad (79)$$

where μ_{IS} , σ_{IS} and δ_{IS} are, respectively, the mean value, the standard deviation and the maximum normalized absolute deviation of the smoothness indicators in the N cells.

For smooth and stable solutions, as shown in Fig. 4, the smoothness indicators $IS_{I_i}^{J, \lfloor \frac{J-1}{2} \rfloor}$ in the N cells, behave as if they were passively advected at the constant speed λ . Consequently, the introduced pseudo-statistical quantities, which satisfy the translation invariance property, remain rather constant as time evolves. This fact represents a useful tool in assessing the stability limits of non-linear ADER-TR schemes. The appearance of destabilizing oscillations within the numerical solution can indeed be detected through an unexpected change in these derived pseudo-statistical quantities; especially in the value of the maximum normalized absolute deviation. Therefore, to establish whether the scheme is stable or not, a fixed threshold on the value of δ_{IS} can be properly tuned; we remark that the value of the threshold obviously depends on the considered initial data.

The main advantage of using the improved indicators of Jiang and Shu (78), is represented by their high sensitivity to small perturbations of the originally smooth profile; this feature makes them capable of detecting even the slightest oscillations inside the numerical

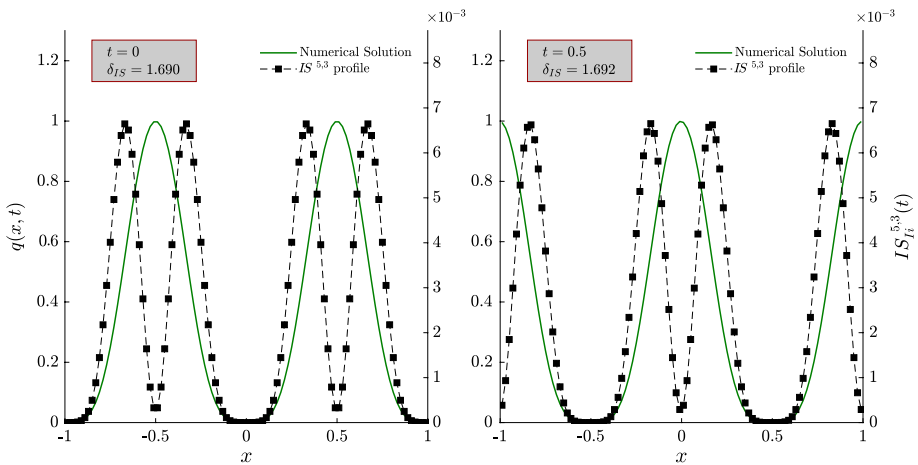


Fig. 4 Stable and smooth numerical solution for the fifth-order ADER-TR1 scheme. The smoothness indicators $IS_{ti}^{5,3}$ behave as if they were advected with the constant propagation speed λ . As a consequence, the values of the derived pseudo-statistical quantities are rather constant as time evolves. See, for example, the value of the maximum normalized absolute deviation δ_{IS} at $t = 0$ (left frame) and $t = 0.5$ (right frame)

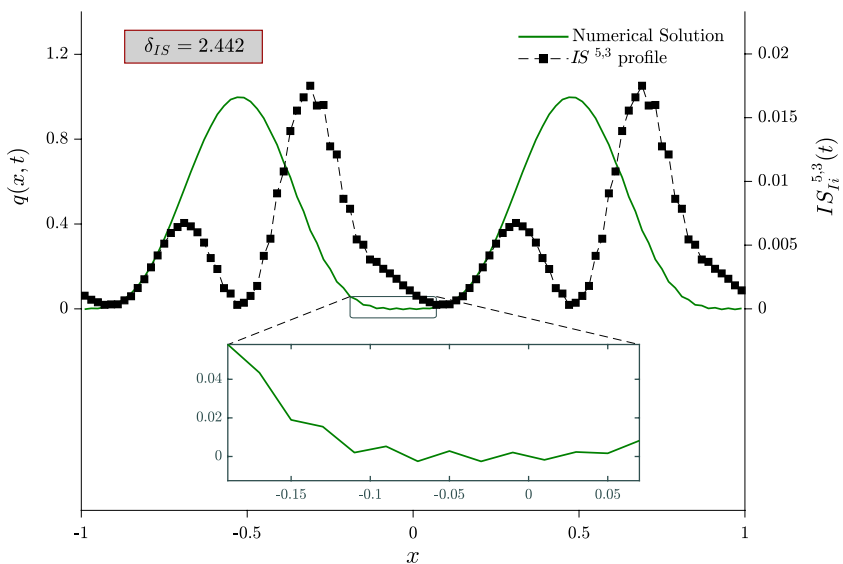


Fig. 5 Behaviour of the smoothness indicators in the presence of small amplitude oscillations in the numerical solution. The profile is highly distorted. Compare with Fig. 4. Moreover, the value of δ_{IS} is notably changed

solution. An example is shown in Fig. 5. It is seen that the regular profile of the smoothness indicators observable in Fig. 4 is profoundly distorted in the presence of small amplitude oscillations. It follows that the value of δ_{IS} is significantly changed, confirming the leading role of the smoothness indicators inside our numerical experiments.

Table 2 Linear stability limits for non-linear ADER-TR schemes up to tenth-order of accuracy. For comparison, it is also listed the approximate CFL condition given by [15] for discontinuous Galerkin methods. Again, the stability limit of the second-order ADER-TR1 and ADER-TR1W schemes are not reported since they boil down to the second-order ADER-TT scheme

J	TR0	TR1	TR0W	TR1W	DG
2	0.59	–	0.42	–	1/3
3	0.38	0.76	0.38	0.76	1/5
4	0.22	0.57	0.22	0.57	1/7
5	0.11	0.44	0.11	0.44	1/9
6	0.06	0.33	0.06	0.33	1/11
7	0.02	0.24	0.02	0.24	1/13
8	0.01	0.18	0.01	0.18	1/15
9	0.007	0.13	0.007	0.13	1/17
10	0.003	0.09	0.003	0.09	1/19

The results of the empirical stability study are shown in Table 2, where schemes of second- up to tenth-order of accuracy are considered. For comparison, we also list the maximum allowable Courant number for explicit discontinuous Galerkin schemes. The CFL stability condition [15] for these methods can be properly expressed in terms of the order of accuracy J as

$$\Delta t_{\text{DG}} \leq \frac{1}{2J-1} \frac{\Delta x}{\lambda}. \quad (80)$$

Notice that the DG schemes considered here correspond to the $P_N P_M$ schemes discussed in [5, 6], with $N = M$. By inspection of the results, some observations are in order:

- i. For non-linear ADER-TR schemes the stability limit decreases monotonically with respect to the order of accuracy.
- ii. The non-linear time-reconstruction procedure does not add any benefit in terms of stability properties. It is only the non-linearity introduced by the spatial reconstruction that has a profound influence on the stability of the schemes.
- iii. Even though ADER-TR schemes have reduced stability regions as compared to standard ADER methods, one sees that the linear stability restrictions are not so prohibitive, especially for ADER-TR1 and ADER-TR1W schemes. Furthermore, we observe that, as already highlighted by the von Neumann analysis, ADER schemes based on the TR1 and TR1W solvers possess improved stability properties, with stability conditions that are much less restrictive than those for ADER-TR0 and ADER-TR0W schemes.
- iv. The CFL conditions for ADER-TR1 and ADER-TR1W schemes are less restrictive than those for DG methods, at least for orders of accuracy lower than tenth. These results give a chance to successfully apply the time-reconstruction approach to ADER methods in the framework of discontinuous Galerkin finite elements without an extra restriction on the time step size.

We finalise the discussion by recalling that for non-linear hyperbolic equations, the CFL-like condition on the time step reads

$$\Delta t \leq \frac{\text{CFL} \Delta x}{S_{\max}^n}, \quad (81)$$

where $S_{\max}^n$ denotes the maximum wave velocity present throughout the domain at time t^n . In practice, for non-linear problems, one heavily relies on linear analysis and on numerical experimentation as guidance. The linear stability condition is a reliable indicator of stability for the non-linear case provided:

- i. the wave speed estimates are themselves reliable and
- ii. a safety coefficient, typically of about 90% of the limiting CFL number arising from the linear stability condition (see Table 2), is used.

4 Numerical Results

In this section we present some results obtained with high-order ADER-TR schemes for scalar hyperbolic equations in one space dimension. In particular, we show the results of our numerical experiments for the fifth- and seventh-order ADER-TR1W schemes. A uniform mesh with N cells is used for all the test cases and the CFL number is always taken as the 90% of the maximum limit on the Courant number imposed by stability constraints. See Table 2. We remark that the main motivation is to demonstrate the capability of simplified ADER-TR schemes to preserve the high accuracy that characterises the ADER approach, maintaining at the same time stable non-oscillatory and sharp discontinuity transitions.

4.1 Convergence Rates

First, we study numerically the convergence properties of ADER-TR schemes and compare them with those of conventional ADER-TT schemes, which use the well-known Toro–Titarev GRP solver [29, 33, 37].

Example 1 We solve the scalar linear advection equation

$$\partial_t q + \lambda \partial_x q = 0 \quad (82)$$

with $\lambda = 1$ in a domain $[-1, 1]$ and a smooth initial condition given by $q_0(x) = \sin^4(\pi x)$ with periodic boundary conditions. The error is measured at $t = 1$. Table 3 shows convergence rates and errors for ADER schemes of fifth- and seventh-order of accuracy. We observe that ADER-TR schemes reach the theoretically expected accuracy. It is interesting to note that the schemes actually well exceed the designed order of accuracy.

Example 2 We solve the inviscid homogeneous Burgers' equation

$$\partial_t q + \partial_x \left(\frac{1}{2} q^2 \right) = 0 \quad (83)$$

with the initial condition $q_0(x) = 0.25 + 0.5\sin(\pi x)$, defined on $[-1, 1]$. Periodic boundary conditions are imposed and the error is measured at time $t = 0.2$. Table 4 shows the numerical results. We can again observe that ADER-TR schemes achieve the expected orders of accuracy with comparable errors for the same mesh. Table 4 shows also convergence rates for ADER schemes with the usual Toro–Titarev GRP solver [29, 33, 37]. Comparing

Table 3 Convergence rates for the linear advection equation. IC: $q_0(x) = \sin^4(\pi x)$. Output time $t = 1$. N is the number of computing cells. For ADER-TT schemes we use a Courant number coefficient CFL = 0.95

Method	N	L_∞ error	L_∞ order	L_1 error	L_1 order
ADER5-TR1W	20	2.08×10^{-2}		2.16×10^{-2}	
	40	7.83×10^{-4}	4.73	4.66×10^{-4}	5.53
	80	3.59×10^{-6}	7.77	1.38×10^{-6}	8.40
	160	1.27×10^{-8}	8.14	1.60×10^{-8}	6.43
	320	4.93×10^{-10}	4.96	4.09×10^{-10}	5.02
ADER5-TT	20	2.00×10^{-3}		1.99×10^{-3}	
	40	1.99×10^{-4}	3.33	5.81×10^{-5}	5.10
	80	7.58×10^{-7}	8.04	4.92×10^{-7}	6.89
	160	1.06×10^{-8}	6.16	1.27×10^{-8}	5.28
	320	3.33×10^{-10}	4.99	4.00×10^{-10}	4.99
ADER7-TR1W	25	1.08×10^{-3}		7.47×10^{-4}	
	50	1.84×10^{-6}	9.20	6.81×10^{-7}	10.10
	100	2.66×10^{-10}	12.75	2.22×10^{-10}	11.59
	200	1.15×10^{-12}	7.86	1.40×10^{-12}	7.30
ADER7-TT	25	1.30×10^{-4}		6.33×10^{-5}	
	50	2.10×10^{-7}	9.27	1.03×10^{-7}	9.26
	100	3.35×10^{-10}	9.30	4.23×10^{-10}	7.93
	200	2.65×10^{-12}	6.98	3.33×10^{-12}	6.99

Table 4 Convergence rates for the Burgers' equation. IC: $q_0(x) = 0.25 + 0.5\sin(\pi x)$. Output time $t = 0.2$. N is the number of computing cells. For ADER-TT schemes we use a Courant number coefficient CFL = 0.95

Method	N	L_∞ error	L_∞ order	L_1 error	L_1 order
ADER5-TR1W	20	7.35×10^{-4}		8.75×10^{-4}	
	40	7.06×10^{-6}	6.70	1.49×10^{-6}	9.20
	80	1.27×10^{-7}	5.80	2.32×10^{-8}	6.00
	160	2.64×10^{-9}	5.58	8.21×10^{-10}	4.82
ADER5-TT	20	7.49×10^{-5}		3.26×10^{-5}	
	40	5.08×10^{-6}	3.88	1.79×10^{-6}	4.19
	80	1.72×10^{-7}	4.88	5.82×10^{-8}	4.94
	160	5.82×10^{-9}	4.89	1.86×10^{-9}	4.96
ADER7-TR1W	20	6.74×10^{-6}		2.14×10^{-6}	
	40	2.21×10^{-8}	8.25	3.69×10^{-9}	9.18
	80	6.22×10^{-11}	8.47	1.71×10^{-11}	7.75
	160	5.79×10^{-13}	6.75	1.66×10^{-13}	6.69
ADER7-TT	20	5.97×10^{-6}		2.42×10^{-6}	
	40	1.10×10^{-7}	5.76	3.64×10^{-8}	6.05
	80	1.09×10^{-9}	6.67	3.18×10^{-10}	6.84
	160	9.17×10^{-12}	6.89	2.55×10^{-12}	6.96

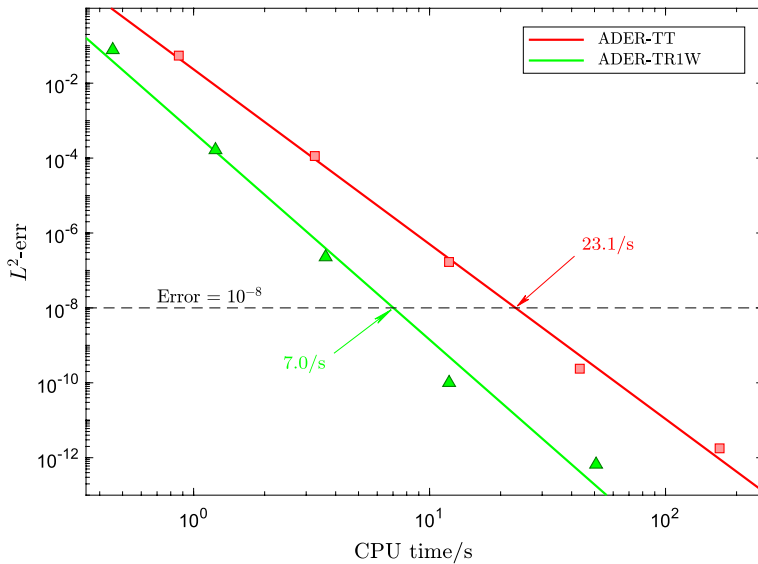


Fig. 6 Efficiency plot: error versus CPU time for a sequence of 5 meshes, for the ideal Euler equations in one space dimension. The computational efficiency of the seventh-order ADER-TR1W and ADER-TT schemes is compared. It is seen that for the same Error = 10^{-8} , the seventh-order ADER-TR scheme is more than three times as fast as the scheme based on the Toro–Titarev GRP solver [33, 37]

the errors, it is evident that the two different approaches give, in terms of accuracy, fairly similar results. It is informative to remark that the small observable deviations in the error values mainly derives from the different CFL coefficient used for advancing the solution in time. For ADER-TT schemes we use a Courant number coefficient $CFL = 0.95$.

4.2 Overview on the Efficiency of the TR Approach

As we have emphasized throughout the paper, the novel time-reconstruction strategy has several advantages, including simplicity and ease of implementation. The present approach is in fact simpler than any of the other approaches that make use of the Cauchy–Kowalewskaya procedure.

In addition, we expect that these inherent features of the time-reconstruction strategy could also lead to an increased efficiency of the resulting ADER methods when applied to complex systems of hyperbolic balance laws. In this case, the fast computation of all time derivatives might indeed outweigh the drawback of a more restrictive CFL condition. Note, however, that for the one-dimensional scalar conservation laws considered in the present paper, the application of the Cauchy–Kowalewskaya procedure is straightforward and therefore no gains can be expected. As a matter of fact, there is a loss of efficiency due to the more restrictive CFL condition of the present ADER-TR schemes.

For the ideal Euler equations, however, our preliminary results indicate that ADER-TR schemes compare very favourably, in terms of efficiency, with ADER methods based on the Toro–Titarev GRP solver [33, 37]. This is mainly the case for the high-order range schemes,

while for the low-order range schemes the standard ADER method remains slightly more efficient due to the enlarged stability limit of unity for all orders.

In Fig. 6 we show an efficiency plot, that is error versus CPU time, for ADER schemes of seventh-order of accuracy for the ideal Euler equations in one space dimension. The plot is constructed by solving the equations on a sequence of meshes with the following initial conditions [38], defined on $[-1, 1]$,

$$p = u = 1, \quad \rho = 2 + \sin^4(\pi x), \quad (84)$$

so that the exact solution is $\rho(x, t) = 2 + \sin^4(\pi(x - t))$, $u = p = 1$. The numerical error in the discrete L^2 norm is recorded at the output time $T_{\text{out}} = 1$ and displayed versus the associated computational cost. Then, a linear least square fitting is suitably deployed to produce the efficiency curves with logarithmic scales on both the horizontal and vertical axes. For a fair comparison between the two schemes, the CFL coefficient is taken as the 90% of the CFL limit determined from the stability analysis. This means that for the ADER-TT scheme we set $\text{CFL} = 0.9$, whereas for the present ADER-TR scheme we use a CFL coefficient equal to 90% of the corresponding value listed in Table 2. A precise comparison is finally made by first fixing an error (horizontal line) and then finding the intercepts with the efficiency curves. For example, for $\text{Error} = 10^{-8}$, Fig. 6 shows that the ADER scheme based on the time-reconstruction approach attains that error more than three time faster than the corresponding ADER-TT scheme. These results, obtained using run times on a single CPU, are representative of the fact that when the application of the Cauchy–Kowalewskaya procedure becomes somewhat cumbersome, the time-reconstruction approach offers advantages not only in terms of ease of implementation but also in terms of efficiency of the schemes.

Finally, we would like to point out that we are eager to address the theme of the efficiency of ADER-TR schemes in a more consistent and systematic way in a future communication dealing with the use of these methods for non-linear systems of hyperbolic balance laws.

4.3 Additional Computational Examples

Example 3 We solve again the scalar linear advection equation (82) with the following initial condition [12, 24]:

$$q(x, 0) = \begin{cases} \exp(-\log(2)(x + 0.7)^2/0.0009), & -0.8 \leq x \leq -0.6, \\ 1, & -0.4 \leq x \leq -0.2, \\ \frac{1 - |10x - 1|}{\sqrt{1 - 100(x - 0.5)^2}}, & 0 \leq x \leq 0.2, \\ 0, & 0.4 \leq x \leq 0.6, \\ 0, & \text{otherwise.} \end{cases} \quad (85)$$

This initial condition consists of a discontinuous square pulse and several continuous but narrow profiles. We use a baseline mesh of 200 cells and periodic boundary conditions. We compute the numerical solution at the output time $t = 8$. Figure 7 shows the solution obtained with the ADER5-TR1W scheme. The solid line corresponds to the exact solution and symbols correspond to the numerical solution. We observe that our simplified ADER-TR scheme is capable of well resolving the sharp peaks in the advected profile.

Example 4 We solve the same non-linear Burgers' equation (83) as that in Example 2 with the following initial condition:

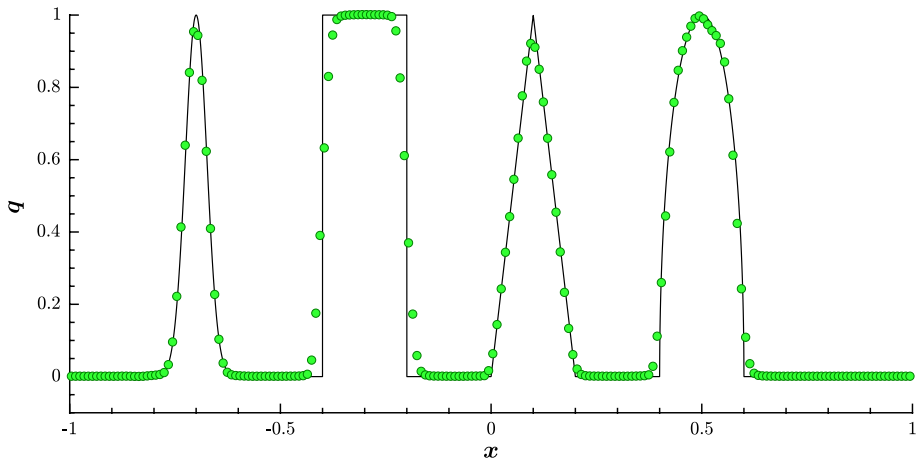


Fig. 7 Computed (symbol) and exact (solid line) solutions for the linear advection equation (82) with initial condition (85) at output time $t = 8$. The fifth-order ADER-TR1W scheme is used on a mesh of $N = 200$ computing cells

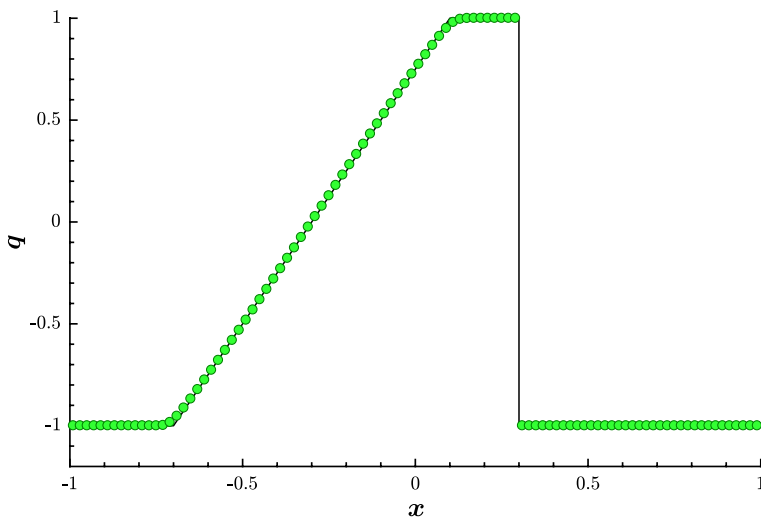


Fig. 8 Homogeneous Burgers' equation with initial condition given by (86). The numerical solution (green circles) computed at $t = 0.4$ on a mesh of 100 cells is compared with the exact solution (solid line). The ADER5-TR1W scheme is used

$$q(x, 0) = \begin{cases} -1, & |x| \leq 0.3, \\ 1, & \text{otherwise.} \end{cases} \quad (86)$$

The exact solution for this problem contains a rarefaction and a stationary shock wave. In Fig. 8 the solution of the ADER5-TR1W scheme on a mesh of $N = 100$ cells is shown. The solid line is the exact solution. We can see that the scheme gives a very satisfactory non-oscillatory shock transition for this problem.

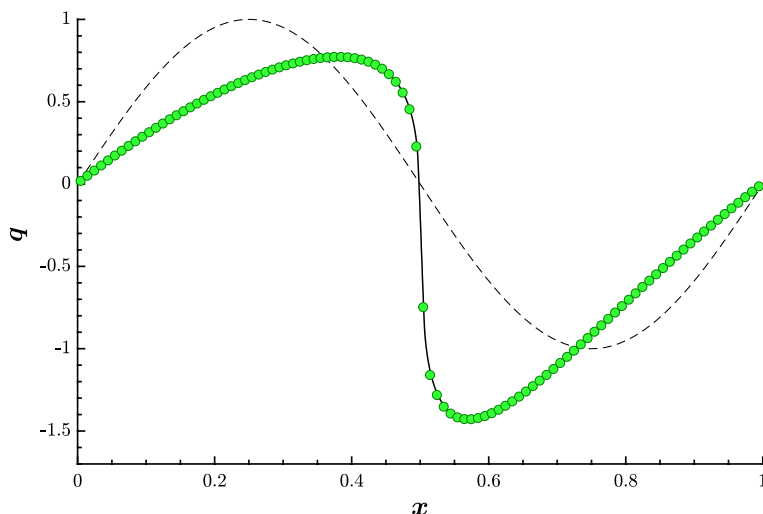


Fig. 9 Burgers' equation with quadratic source term ($\beta = -2$) and the initial condition (dash line) $q_0(x) = \sin(2\pi x)$. The numerical solution (green circles) computed at $t = 0.15$ on a mesh of 100 cells is compared with the exact solution (solid line). The ADER5-TR1W scheme is used

Example 5 This example considers the numerical solution of the inviscid Burgers' equation with a quadratic source term

$$\begin{cases} \partial_t q + \partial_x \left(\frac{1}{2} q^2 \right) = \beta q^2, \\ q(x, 0) = q_0(x). \end{cases} \quad (87)$$

By following the characteristic curve method the solution of (87) is obtained as [21]

$$q(x, t) = \frac{q_0(y)}{1 - \beta t q_0(y)} \quad (88)$$

with y satisfying

$$x = y - \frac{\log(1 - \beta t q_0(y))}{\beta}. \quad (89)$$

We compute numerically the solution of (87) for $\beta = -2$ and the initial condition $q_0(x) = \sin(2\pi x)$, defined on $[0, 1]$. Periodic boundary conditions are imposed. In Fig. 9 we compare the numerical solution of the ADER5-TR1W scheme with the exact solution given by (88) at the output time $t = 0.15$. A uniform mesh of $N = 100$ cells is used.

Example 6 We solve the generalized Riemann problem for the inviscid inhomogeneous Burgers' equation:

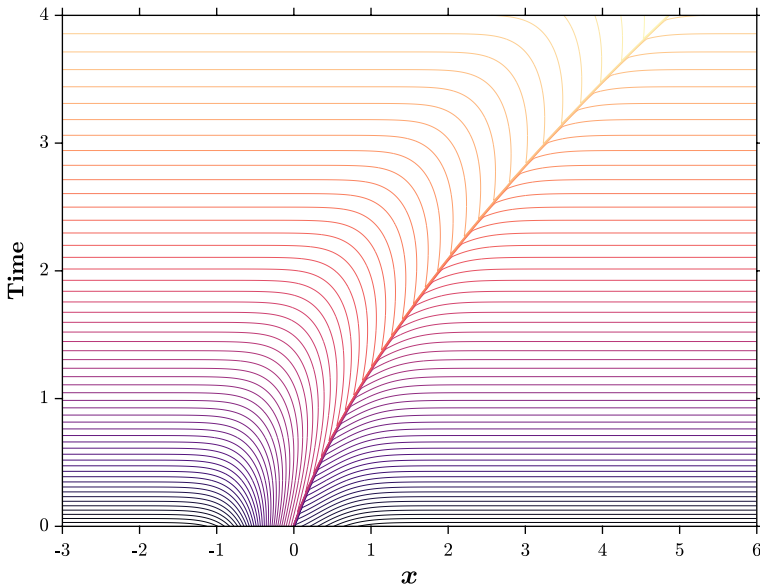


Fig. 10 Numerical solution of the GRP problem (90) in the $x - t$ plane. The ADER5-TR1W scheme on a mesh of 800 cells is used. Compare with the results reported in [28]

$$\begin{cases} \partial_t q + \partial_x \left(\frac{1}{2} q^2 \right) = e^{-q}, \\ q(x, 0) = \begin{cases} q_L(x) = e^{-2\left(x - \frac{1}{5}\right)^2} & \text{if } x < 0, \\ q_R(x) = \frac{1}{4} e^{-2\left(x + \frac{1}{5}\right)^2} & \text{if } x > 0. \end{cases} \end{cases} \quad (90)$$

Figure 10 shows the numerical solution in the $x - t$ plane computed by the ADER5-TR1W scheme on a fine mesh of 800 cells. The dominant feature of the solution is an accelerating shock wave that emerges from the initial discontinuity in the initial condition at $x = 0$. We observe that the scheme gives very satisfactory results, properly accounting for the presence of a non-linear source term. Compare with the results reported in [28].

Example 7 We finally test the fifth-order ADER-TR1W scheme on the non-linear non-convex scalar Buckley–Leverett problem

$$\begin{cases} \partial_t q + \partial_x \left(\frac{4q^2}{4q^2 + (1 - q)^2} \right) = 0, \\ q(x, 0) = \begin{cases} 1 & \text{when } -\frac{1}{2} \leq x \leq 0, \\ 0 & \text{elsewhere.} \end{cases} \end{cases} \quad (91)$$

The solution is computed up to $t = 0.4$. The exact solution is a shock–rarefaction–contact discontinuity mixture. In Fig. 11 the numerical solution of the ADER5-TR1W scheme on a mesh of 100 cells is shown. One notes that the scheme is able to provide a good resolution of the correct entropy solution for this problem.

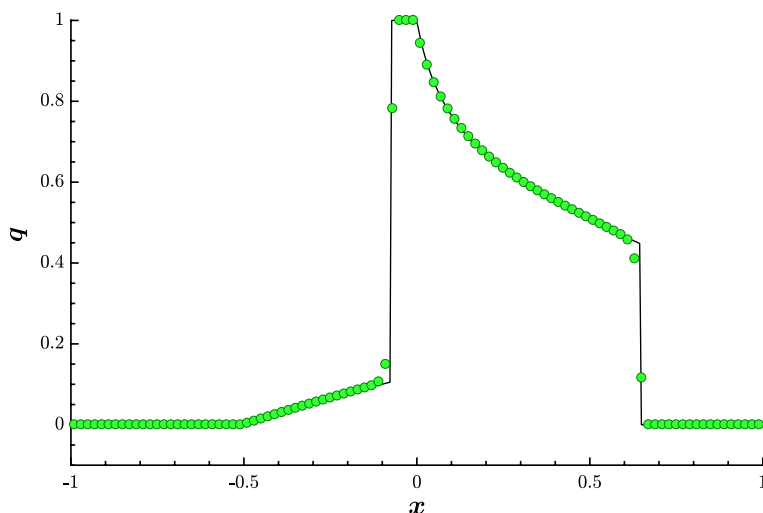


Fig. 11 The Buckley–Leverett problem. Numerical solution (green circles) computed at $t = 0.4$ with 100 cells using the ADER5-TR1W scheme. Solid line: exact solution

5 Conclusions

A new method to compute the solution of the generalized Riemann problem GRP_K at the element interface has been presented, which avoids the cumbersome Cauchy–Kowalewskaya procedure used in the original GRP_K [37]; see also [3, 33]. The novel approach relies on a time-reconstruction procedure. A careful stability analysis of the resulting numerical methods for both linear and non-linear schemes has been carried out; the linear stability limitations on the Courant number for ADER-TR schemes up to tenth-order of accuracy have been assessed. We found that unlike the standard ADER methods [37], the simplified finite-volume ADER schemes based on the new family of GRP_K solvers do not have optimal stability condition. This limitation is offset by the simplifications of the scheme, ease of implementation and efficiency.

Numerical results clearly indicate that for the 1D scalar problems the schemes compare favourably with conventional ADER methods. Indeed, for smooth solutions the high accuracy is retained, whereas for discontinuous solutions good essentially non-oscillatory results and sharp resolution of discontinuities are satisfactorily maintained. Preliminary results for the one-dimensional Euler equations suggest that the new time-reconstruction approach offers advantages both in terms of ease of implementation and efficiency of the methods. Moreover, we expect that such benefits will be more significant for very complicated non-linear multidimensional problems, the subject of ongoing research. Forthcoming research will also explore the advantages of a Hermite-type interpolation in the construction of ADER-TR1 schemes and will investigate the applicability of the time-reconstruction strategy to ADER schemes in the framework of the discontinuous Galerkin (DG) finite element methods.

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